

27 July 2026

Melanoma Detection

from Dermoscopic Images

GROUP 15

E/22/052 , E/22/058, E/22/353/, E/22/419



Dataset Status and Source

Source

- Melanoma Cancer Image Dataset, Kaggle (bhaveshmittal)
- ~10,000 dermoscopic lesion images, benign vs. malignant
- Pre-sized 224×224 RGB, existing train/test partitions

Privacy

- Public, de-identified images
no patient records, minimal privacy risk.

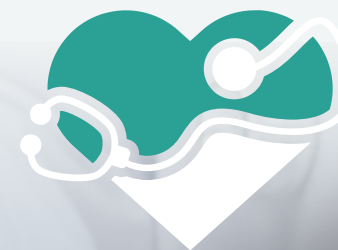
STATUS

Dataset downloaded and loaded

- Downloaded via the Kaggle API into the training environment
- Loaded successfully with PyTorch's ImageFolder; class names and counts confirmed
- Train/validation split (85/15) carved from the provided training partition

Known limitations

- Class imbalance between benign/malignant
- Variable image quality across sources
- Limited demographic diversity



Preprocessing and Annotation Plan

1. Labels

Ground-truth benign/malignant labels are provided with the dataset, no manual annotation required.

2. Normalize

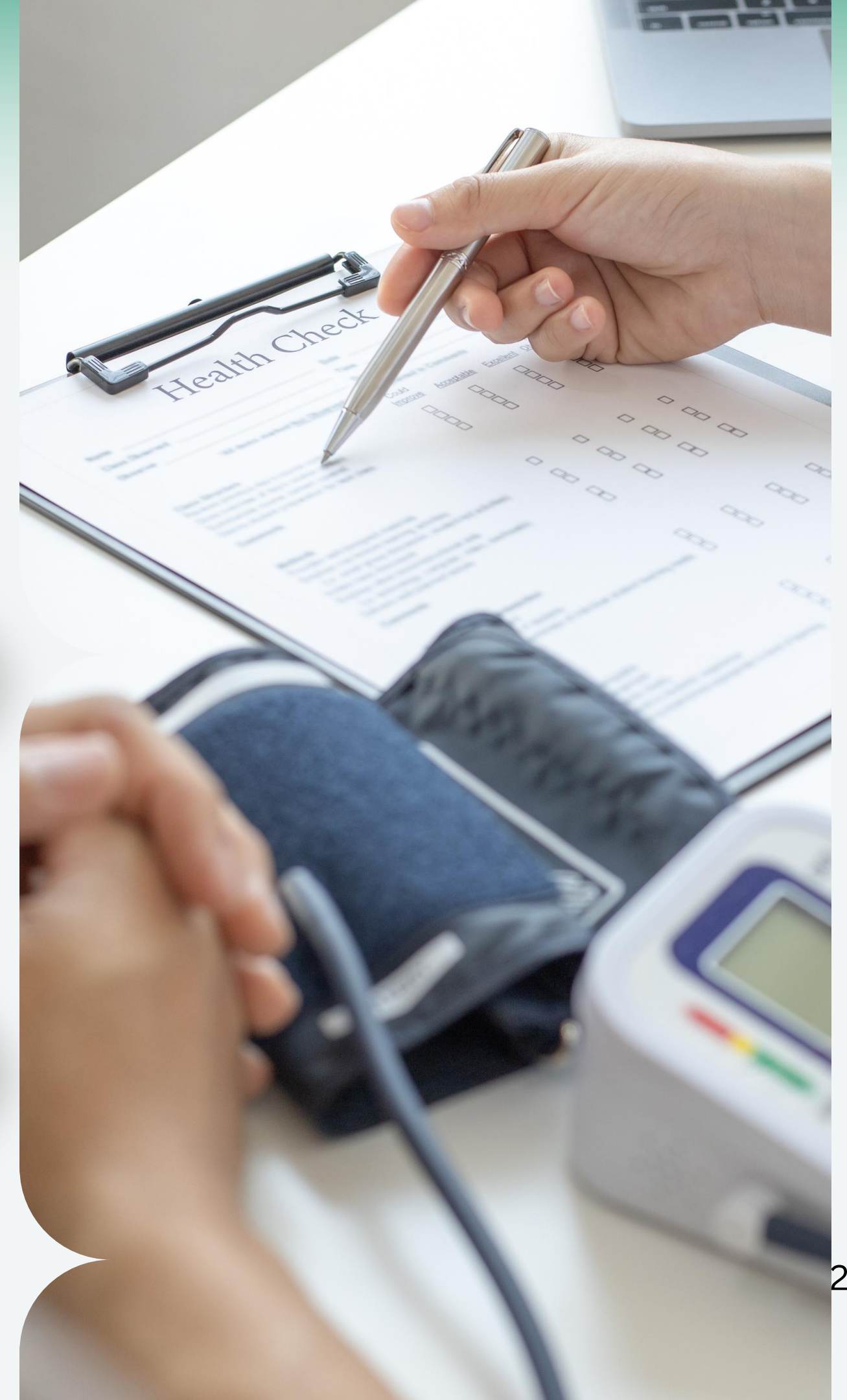
Pixel values scaled and normalized to ImageNet statistics to match the pretrained backbone.

3. Split

85/15 train/validation split carved from the training partition; original test set held out untouched.

4. Augment

Random flips, rotations, and color jitter applied to the training split only, to offset the limited dataset size.



Baseline Pipeline & Methodology

Evaluation metrics

- Accuracy
- Precision
- Sensitivity / Recall (F1)
- Confusion matrix
- Saliency visualization for explainability

Baseline

Pretrained ResNet18/MobileNetV2 as a frozen feature extractor with a linear classification head. Establishes the lower bound.

Primary model

Fine-tuned pretrained convolutional backbone, adapted to binary classification with the last block unfrozen and a low learning rate.



Initial Implementation Results

Status: baseline model (EfficientNetV2-S) trained end to end with early stopping. Below are validation-set metrics from the completed run.

Accuracy

Final: 94.0%

Precision

91.2%

Sensitivity

97.4%

F1-score

0.942

Qualitative observations

- Sensitivity (97.4%) trails precision (91.2%) : correctly identifies 974 of 1000 malignant cases, only 26 false negatives, the exact failure mode flagged as highest-cost in the proposal
- Early stopping halted training on the loss trend, but evaluation ran on the final epoch's weights rather than the best one (epoch 8, val loss 0.1452, accuracy 95.25%) , restoring the best checkpoint is a quick, high-value fix

Current Challenges & Next Steps

Challenges

Class imbalance

Benign/malignant ratio not yet quantified; will drive the choice between class-weighted loss and minority-class augmentation.

Overfitting risk

Moderate dataset size relative to backbone capacity, mitigated via transfer learning, augmentation, and early stopping.

Spurious correlations

Ruler marks, ink markings, and hair visible in some images risk being learned as shortcut features.

Next steps

This week

Finish dataset download, run class-balance audit, execute baseline (frozen backbone) end to end.

By 18 Aug

Fine-tuned model functional with preliminary quantitative + qualitative results.

By 25 Aug

Freeze experiments; shift to ablations and failure-case analysis.

By 1 Sep

Draft report ready ahead of 7 Sep submission and demo.

 Repository: github.com/cepdnack/e22-co5430-melanoma-cancer-detection